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In [1]: import numpy as np
import matplotlib.pyplot as plt
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Exercise 6.1.1

For each IVP, determine whether the problem satisfies the conditions of Theom 6.1.2. If so, determine the smallest possible value for L .

(a) $f(t, u) = 3u, 0 \leq t \leq 1$

$$\frac{\partial f}{\partial u} = 3$$
$$|3| \leq 3 = L$$

(b) $f(t, u) = -t \sin(u), 0 \leq t \leq 5$

$$\frac{\partial f}{\partial u} = -t \cos(u)$$
$$|-t \cos(u)| = t \cos(u) \leq 5 \cos(0) = 5 = L$$

(c) $f(t, u) = -(1 + t^2)u^2, 1 \leq t \leq 3$

$$\frac{\partial f}{\partial u} = -2(1 + t^2)u$$

This function is not bounded by a constant L since the u term can scale $|-2(1 + t^2)|$ arbitrarily large.

(d) $f(t, u) = \sqrt{u}, 0 \leq t \leq 1$

$$\frac{\partial f}{\partial u} = \frac{1}{2}u^{-1/2}$$

This function is not bounded by a constant L since $u = 0$ blows up to infinity.

Exercise 6.1.4

Consider the IVP $u' = u^2, u(0) = \alpha$.

(a) Does Theorem 6.1.1 apply to this problem?

Theorem 6.1.1 does not apply to this problem. $\partial f / \partial u = 2u$, which isn't bounded by a constant L since u can be arbitrarily large.

(b) Show that $u(t) = \alpha / (1 - \alpha t)$ is a solution of the IVP.

$$\begin{aligned} u'(t) &= (\alpha(1 - \alpha t)^{-1})' \\ &= -\alpha(1 - \alpha t)^{-2} \cdot (-\alpha) \\ &= \alpha^2(1 - \alpha t)^{-2} \\ &= (\alpha / (1 - \alpha t))^2 \\ &= u^2 \end{aligned}$$

(c) Does this solution necessarily exist for all $t \in [0, 1]$?

Yes, since there is no t term in $u(t)$ that could create a discontinuity.

Exercise 6.2.1

Do two steps of Euler's method for the following problems using the given step size h . Then, compute the error using the given exact solution.

(a) $u' = -2tu$, $u(0) = 2$; $h = 0.1$; $\hat{u}(t) = 2e^{-t^2}$

$$\begin{aligned} u_1 &= u_0 + hf(t_0, u_0) = 2 + 0.1(-2(0)(2)) = 2 \\ u_2 &= u_1 + hf(t_0 + h, u_1) = 2 + 0.1(-2(0.1)(2)) = 1.96 \\ \hat{u}(t_0 + 2h) &= 1.921578878 \\ |1.921578878 - 1.96| &= 0.0384211217 \end{aligned}$$

(b) $u' = u + t$, $u(0) = 2$; $h = 0.2$; $\hat{u}(t) = -1 - t + 3e^t$

$$\begin{aligned} u_1 &= u_0 + hf(t_0, u_0) = 2 + 0.2(2 + 0) = 2.4 \\ u_2 &= u_1 + hf(t_0 + h, u_1) = 2.4 + 0.2(2.4 + 0.2) = 2.92 \\ \hat{u}(t_0 + 2h) &= 3.075474093 \\ |3.075474093 - 2.92| &= 0.1554740929 \end{aligned}$$

$$(c) \, tu' + u = 1, u(1) = 6; h = 0.25; \hat{u}(t) = 1 + 5/t$$

$$u' = (1 - u)/t$$

$$u_1 = u_0 + hf(t_0, u_0) = 6 + 0.25((1 - 6)/1) = 4.75$$

$$u_2 = u_1 + hf(t_0 + h, u_1) = 4.75 + 0.25((1 - 4.75)/0.25) = 1$$

$$\hat{u}(t_0 + 2h) = 11$$

$$|11 - 1| = 10$$

$$(d) \, u' - 2u(1 - u) = 0, u(0) = 1/2; h = 0.25; \hat{u}(t) = 1/(1 + e^{-2t})$$

$$u' = 2u(1 - u)$$

$$u_1 = u_0 + hf(t_0, u_0) = 0.5 + 0.25(2(0.5)(1 - 0.5)) = 0.625$$

$$u_2 = u_1 + hf(t_0 + h, u_1) = 0.625 + 0.25(2(0.625)(1 - 0.625)) = 0.7421875$$

$$\hat{u}(t_0 + 2h) = 0.7310585786$$

$$|0.7310585786 - 0.7421875| = 0.0111289214$$

Exercise 6.3.1

Rewrite the given higher order problems as first-order systems.

$$(a) \, y''' - 3y'' + 3y' - y = t, y(0) = 1, y'(0) = 2, y''(0) = 3$$

Where $u_i = y^{(i-1)}$,

$$u'_1 = u_2$$

$$u'_2 = u_3$$

$$u'_3 = t + u_1 - 3u_2 + 3u_3$$

$$u_1(0) = 1, u_2(0) = 2, u_3(0) = 3$$

$$(b) \, y'' + 4(x^2 - 1)y' + y = 0, y(0) = 2, y'(0) = -1$$

Where $u_i = y^{(i-1)}$,

$$\begin{aligned}u_1' &= u_2 \\u_2' &= -u_1 - 4(x^2 - 1)u_2 \\u_1(0) &= 2, u_2(0) = -1\end{aligned}$$

Exercise 6.3.2

Write the given IVP as a system. Then do two steps of Euler's method by hand (perhaps with a calculator) with the indicated step size h . Using the given exact solution, compute the error after the second step.

(a) $y'' + 4y = 4t, y(0) = 1, y'(0) = 1; \hat{y}(t) = t + \cos(2t), h = 0.1$

Where $u_i = y^{(i-1)}$,

$$\begin{aligned}u_1' &= u_2 \\u_2' &= 4t - 4u_1 \\u_1(0) &= 1, u_2(0) = 1\end{aligned}$$

$$\begin{aligned}\begin{bmatrix} u_1 \\ u_2 \end{bmatrix}_1 &= \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}_0 + h \begin{bmatrix} f_1(t_0, u_{1,0}) \\ f_2(t_0, u_{2,0}) \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} + 0.1 \begin{bmatrix} 1 \\ 4(0) - 4(1) \end{bmatrix} = \begin{bmatrix} 1.1 \\ 0.6 \end{bmatrix} \\ \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}_2 &= \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}_1 + h \begin{bmatrix} f_1(t_0 + h, u_{1,1}) \\ f_2(t_0 + h, u_{2,1}) \end{bmatrix} = \begin{bmatrix} 1.1 \\ 0.6 \end{bmatrix} + 0.1 \begin{bmatrix} 1.1 \\ 4(0.1) - 4(1.1) \end{bmatrix} = \begin{bmatrix} 1.21 \\ 0.2 \end{bmatrix} \\ \hat{y}(t_0 + 2h) &= 1.121060994 \\ |1.121060994 - 1.21| &= 0.088939006\end{aligned}$$

Exercise 6.4.1

For each IVP, write out (possibly using a calculator) the first time step of the improved Euler method with $h = 0.2$.

(a) $u' = -2tu, 0 \leq t \leq 2, u(0) = 2; \hat{u}(t) = 2e^{-t^2}$

$$\begin{aligned}u_1 &= u_0 + hf(t_0 + (1/2)h, u_0 + (1/2)hf(t_0, u_0)) \\&= 2 + 0.2f(0 + (1/2)(0.2), 2 + (1/2)(0.2)(-2(0)(2))) \\&= 2 + 0.2f(0.1, 2) \\&= 2 + 0.2(-2(0.1)(2)) \\&= 1.92\end{aligned}$$

Exercise 6.5.1

Using Function 6.5.2 with an error tolerance of 10^{-8} , solve $y'' + (1 + y')^3 y = 0$ over $0 \leq t \leq 4\pi$ with the indicated initial conditions. Plot $y(t)$ and $y'(t)$ as functions of t and separately plot the time step size as a function of t .

```
In [2]: def rk23(du_dt, tspan, u0, tol):
        """
        rk23(du_dt, tspan, u0, tol)

        Apply adaptive embedded RK formula to solve the vector-valued IVP u'=du_dt(u,p,t)
        over the interval tspan with u(tspan[1])=u0, with error tolerance tol.
        Return a vector of times and a vector of solution values/vectors.
        """

        # Initialize for the first time step.
        t = [tspan[0]]
        u = np.array([np.atleast_1d(u0)])
        i = 0
        h = 0.5 * tol ** (1 / 3)
        s1 = du_dt(t[0], u[0])

        # Time stepping.
        while t[i] < tspan[-1]:
            # Detect underflow of the step size.
            if t[i] + h == t[i]:
                print(f"Stepsize too small near t={t[i]}")
                break # quit time stepping loop

            # New RK stages.
            s2 = du_dt(t[i] + h / 2, u[i] + (h / 2) * s1)
            s3 = du_dt(t[i] + 3 * h / 4, u[i] + (3 * h / 4) * s2)
            unew2 = u[i] + h * (2 * s1 + 3 * s2 + 4 * s3) / 9 # 2rd order solution
            s4 = du_dt(t[i] + h, unew2)
            err = h * (-5 * s1 / 72 + s2 / 12 + s3 / 9 - s4 / 8) # 2nd/3rd order difference
            E = np.linalg.norm(err, np.inf) # error estimate
            maxerr = tol * (1 + np.linalg.norm(u[i], np.inf)) # relative/absolute blend

            # Accept the proposed step?
            if E < maxerr: # yes
                t.append(t[i] + h)
                u = np.vstack([u, unew2])
                i += 1
                s1 = s4 # use FSAL property

            # Adjust step size.
            q = 0.8 * (maxerr / E) ** (1 / 3) # conservative optimal step factor
            q = min(q, 4) # limit stepsize growth
            h = min(q * h, tspan[-1] - t[i]) # don't step past the end

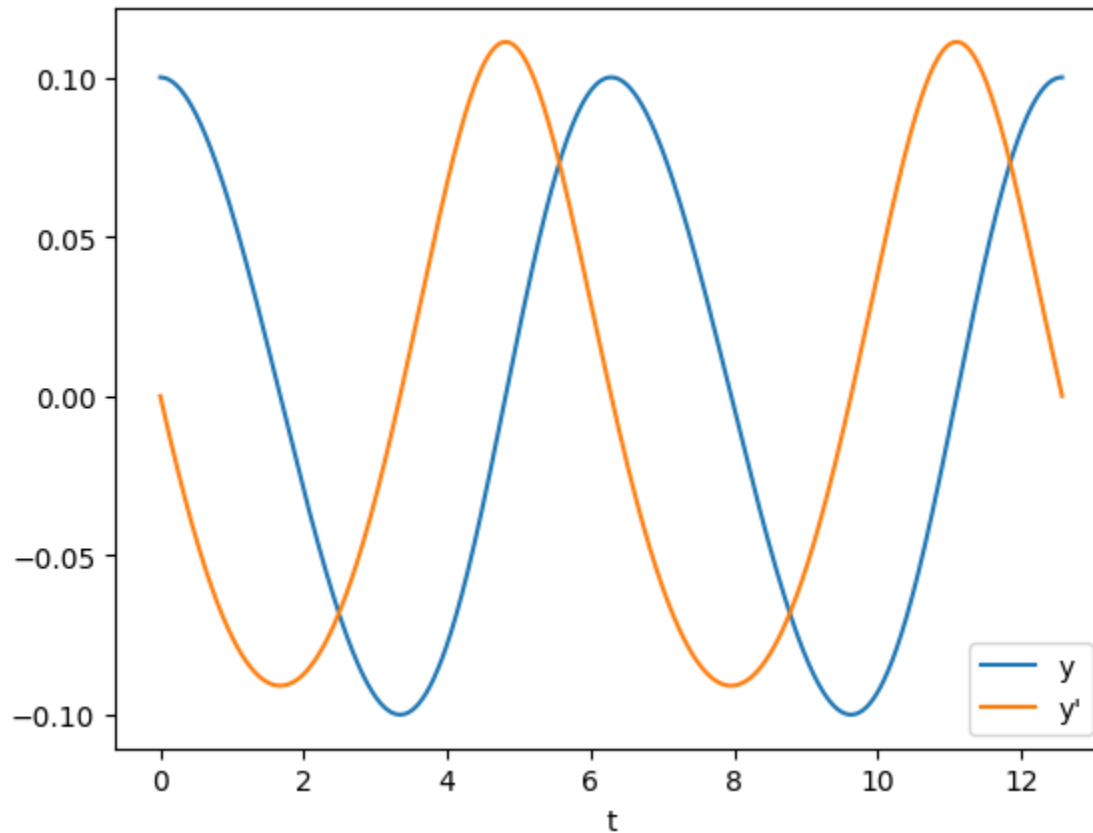
        # Convert outputs to arrays
```

```
return np.array(t), u.T
```

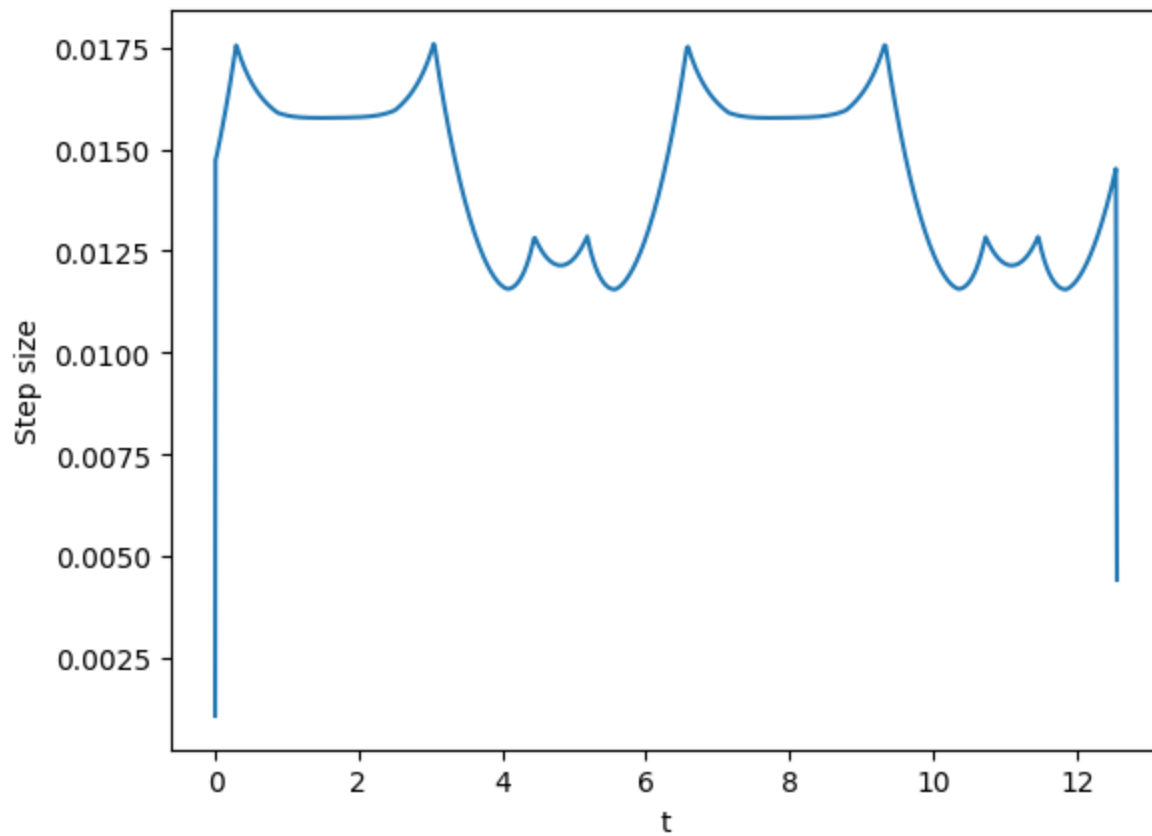
(a) $y(0) = 0.1, y'(0) = 0$

```
In [12]: f = lambda t, u: np.array([u[1], -(1+u[1])**3 * u[0]])
t, u = rk23(f, [0, 4 * np.pi], [0.1, 0], 1e-8)

plt.plot(t, u[0, :], label = "y")
plt.plot(t, u[1, :], label = "y'")
plt.xlabel("t")
plt.legend()
plt.show()
```



```
In [13]: dt = [t[i + 1] - t[i] for i in range(t.size - 1)]
plt.plot(t[:-1], dt)
plt.xlabel("t")
plt.ylabel("Step size")
plt.show()
```



Exercise 6.6.1

For each method, write out the generating polynomials $\rho(z)$ and $\sigma(z)$.

(a) AM2

$$\rho(z) = z - 1, \sigma(z) = \frac{1}{2}z + \frac{1}{2}$$

(b) AB2

$$\rho(z) = z^2 - z, \sigma(z) = \frac{3}{2}z - \frac{1}{2}$$

(c) BD2

$$\rho(z) = z^2 - \frac{4}{3}z + \frac{1}{3}, \sigma(z) = \frac{2}{3}z^2$$

(d) AM3

$$\rho(z) = z^2 - z, \sigma(z) = \frac{5}{12}z^2 + \frac{8}{12}z - \frac{1}{12}$$

(e) AB3

$$\rho(z) = z^3 - z^2, \sigma(z) = \frac{23}{12}z^2 - \frac{16}{12}z + \frac{5}{12}$$

Exercise 6.6.2

Find the solution value u_1 produced by AM1 (backward Euler) for each IVP. (Reminder: This is an implicit formula, so you may need to solve a nonlinear equation. If there are multiple possible values for u_1 , choose the one closest to u_0 .)

(a) $u' = -2tu, 0 \leq t \leq 2, u_0 = 3, h = 0.2$

$$u_{i+1} = u_i + hf_{i+1} = u_i + hf(t_{i+1}, u_{i+1})$$

So

$$\begin{aligned} u_1 &= u_0 + hf(t_1, u_1) \\ &= 3 + (0.2)(-2(0.2)(u_1)) \\ &= 3 - 0.08u_1 \\ \rightarrow 1.08u_1 &= 3 \\ \rightarrow u_1 &= 2.\bar{7} \end{aligned}$$